

Source Integration

- NCERT Class 11 — India: Physical Environment, Chapter 5: Natural Vegetation

Core Factual & Conceptual Architecture

1. Forest Groups and Typologies

- Natural Vegetation Definition: Plant communities left undisturbed over long periods, allowing species to adapt fully to local climate and soil.
- Five Forest Groups:
 - Tropical Evergreen & Semi-Evergreen: Western Ghats, NE hills, and Andaman-Nicobar (>200 cm rainfall, >22°C temp). Stratified canopies featuring rosewood, mahogany, and ebony.
 - Tropical Deciduous (Monsoon Forests): Most widespread type (70–200 cm rainfall). Divided into moist deciduous (teak, sal, shisham) and dry deciduous parklands (tendu, palas).
 - Tropical Thorn Forests: Semi-arid zones (<50 cm rainfall) in Punjab, Haryana, Rajasthan, featuring babool, neem, and cactus scrub.
- Historical Shift: British commercial exploitation replaced native oak with pine for railways and cleared land for tea, coffee, and rubber plantations.

2. Montane and Littoral Ecosystems

- Northern Mountain Forests: Himalayan zonation shifting from foothills deciduous, to 1,000–2,000m wet temperate oak-chestnut, to chir pine, deodar, blue pine, alpine pastures, and tundra mosses/lichens. Transhumance practiced by Gujjars, Bakarwals, Bhotiyas, and Gaddis.
- Southern Mountain Forests: Higher Western Ghats, Vindhyas, and Nilgiris featuring temperate forests known as **sholas**.
- Littoral and Swamp Forests: Wetlands covering 3.9 million hectares across eight national categories. Mangroves cover 4,992 sq. km (7 percent of global total), concentrated in the Sunderbans and Andaman-Nicobar islands.

3. Forest Conservation & Social Forestry

- National Forest Policy: Adopted in 1952 and modified in 1988, targeting 33 percent forest cover, environmental stability, and conservation of natural heritage.
- Social Forestry Categories (National Commission on Agriculture, 1976):
 - Urban Forestry: Green belts and roadside plantings in urban centers.
 - Rural Forestry: Comprises agro-forestry (trees + crops) and community forestry on public land.
 - Farm Forestry: Commercial or non-commercial tree growing by farmers on private land using free departmental seedlings.

4. Wildlife Conservation and Protected Areas

- Decline Drivers: Industrial exploitation, land clearance for agriculture and dams, domestic livestock grazing, and rampant commercial poaching.
- Legislative Framework: Wildlife (Protection) Act of 1972 (amended in 1991) establishes legal backing for 107 National Parks and 573 Wildlife Sanctuaries.
- Biosphere Reserves: 18 reserves integrated into national protection, with 12 recognized under UNESCO's Man and Biosphere (MAB) Programme (e.g., Nilgiri, Sunderban, Nanda Devi, Nokrek).
- Flagship Projects: Project Tiger (launched in 1973, expanding to 58 reserves with 3,682 tigers in 2023) and Project Elephant (launched in 1992).